

Examiner reports

Q1.

This question was very well done, with over 87% of students correctly choosing option 2.

Q2.

There were some good attempts at parts (a) and (b). In (a) students were able to use the binomial distribution function on their calculators to obtain the correct final answer. In (b) students generally recognised the need to cube their answer to (a), however it was disappointing that some multiplied their answer to (a) by 3, obtaining an answer greater than 1.

In part (c) many students did not state their assumptions in context or instead gave 'scientific/practical' rather than 'statistical' assumptions. Examples include: all extraneous variables kept constant, the same darts and dart board are used and the distance from the dart board remains the same each time. It is important that students know the correct distributional assumptions for the binomial distribution and are able to apply them to the context of a question. In this case a 'fixed number of trials' was not given credit as this was stated in the question.

Q3.

The vast majority of students answered this multiple choice question correctly, giving the answer $\frac{1}{10}$.

Q4.

Many students scored full marks on (a), though significant numbers scored 1 out of 2 because their conclusion was not drawn clearly enough. In addition, significant numbers failed to score at all because they did not understand the significance of the probabilities in determining independence.

(b) was a very well answered question, with the majority of students scoring full marks, though there was some confusion about the meaning of "mathematics or biology or both".

Q7.

This fairly standard probability question posed major difficulties to the weaker students whereas the stronger students scored high marks often with an apparent minimum of effort. Almost all students answered part (a)(i) correctly with slightly fewer scoring the mark in part (a)(ii) through the use of multiplication instead of addition. The use of conditional probability to answer parts (a)(iii) to (v) was beyond many students and it was not unusual to see repeated answers of 0.25 or 0.30. Of those students who scored very few marks in part (a), many were then able to make a sensible and often correct attempt at part (b). Whilst many students identified the need for $(0.30^2 + 0.25^2)$, this was sometimes spoilt by the introduction of 0.55, 0.55^{-1} or 2 as a multiplier.

Q8.

Answers to this question on probability were, in the main, of a high standard and showed

a marked improvement on similar questions in previous papers. It was quite rare to see an incorrect answer to part (a) with $0.90 + 0.95 - 0.855$ in (ii) being by far the most common error. Most students also scored well in part (b) with few students not scoring full marks in (i) to (iii). However, (iv) proved more challenging with $0.10 \times 0.15 + 0.05 \times 0$ as a frequent incorrect answer. Even some better students lost 1 mark due to calculating $0.085 + 0.05$ instead of 0.085×0.05 .

Q9.

Almost without exception, full marks were scored in part (a). As a consequence, students usually scored well in part (b) with full marks by far the most common mark. In fact there was no common error but simply one-off methodological or numerical errors. Again in part

(c), many students had the correct fractions of $\frac{0.090}{0.155}$ and $\frac{0.065}{0.155}$ or their equivalents. Where full marks were not scored, it was often due to not including a multiplier of 2.

Q10.

In answering this question, some students ignored the instruction regarding answers 'to three decimal places' and simply left them as fractions. This was not penalised in part (a) but was penalised in part (b). Most students answered parts (a)(i) & (ii) correctly but in part (a)(iii), some incorrectly assumed independence and so multiplied together their previous two answers. Part (a)(iv) caused significantly more students a problem. Whilst many obtained the correct numerator of 153, they used a denominator of 640 instead of 172. Most students scored only 1 or 2 marks in part (b). Whilst some were able to quote the three correct numerators, they then used 194^3 or $(640 \times 639 \times 638)$ in the denominator or failed to multiply by $3! = 6$.

Q11.

In answering this question, a significant proportion of students ignored the instruction regarding answers 'to three decimal places' and simply left them as fractions. This was not penalised in part (a) but was penalised in parts (b) & (c). Most students answered parts (a)(i) & (ii) correctly but in part (a)(iii), many incorrectly assumed independence and so multiplied together their previous two answers. Part (a)(iv) caused significantly more students a problem. Whilst many obtained the correct numerator of 153, they used a denominator of 640 instead of 172. Few students were able to make significant progress in part (b). Many responses lacked any attempt at a numerical justification, whilst others tried to identify new values to use from the table rather than following the instruction (and hint) to use answers from part (a). Such answers usually lacked explanation, left answers as fractions that were difficult to compare, or did not consider like-for-like situations (eg $B = 3$ with $B \leq 3$). Most students scored only 1 or 2 marks in part (c). Whilst most were able to quote the three correct numerators, many used 194^3 or $(640 \times 639 \times 638)$ in the denominator and even more failed to multiply by $3! = 6$. Sadly, it was also not rare to see the three correct fractions added together.

Q12.

Many candidates scored between 5 and 9 marks, with the marks usually being lost in parts (b) and (c). Too many candidates lost valuable marks for ignoring 'to three decimal places'. Fractional answers only scored full marks in parts (a)(i) to (a)(iii) where almost all candidates scored the 3 marks available. In part (a)(iv), the use of $P(L \cap F)$ or $P(F | L)$, rather than $P(L | F)$, was often in evidence. In part (b), far too many candidates used:

$$\left(\frac{94}{126}\right)^2 \text{ or } \frac{94}{126} \times \frac{93}{126} \text{ instead of } \frac{94}{126} \times \frac{93}{125}.$$

In answering part (c), many candidates had the correct numerator of $349 \times 193 \times 103$ but a majority then used a denominator of $(645)^3$ and almost all candidates omitted the permutation multiplier of $3! = 6$.

Q13.

Most candidates scored between 7 and 10 marks, with the marks usually being lost in parts (b) and (c). Too many candidates lost valuable marks for ignoring 'to three decimal places'. Fractional answers only scored full marks in parts (a)(i) to (a)(iii), where almost all candidates scored the 3 marks available. In part (a)(iv), $P(L \cap F)$ or $P(F | L)$, rather than $P(L | F)$, was often found. Similarly, the use of $P(L' \cap M)$ or $P(L' | M)$ was often in evidence in answers to part (a)(v).

In part (b), far too many candidates used $\left(\frac{94}{126}\right)^2$ or $\frac{94}{126} \times \frac{93}{126}$ instead of $\frac{94}{126} \times \frac{93}{125}$.

In answering part (c), many candidates had the correct numerator of $349 \times 193 \times 103 \times$ but a majority then used a denominator of $(645)^3$, and almost all candidates omitted the permutation multiplier of $3! = 6$.

Q14.

This question on the binomial distribution was a good source of marks for many candidates, with the more able often scoring most of the 15 marks available. Save for a small minority of candidates who calculated $P(R = 5)$ using the formula, part (a)(i) was usually answered correctly from the appropriate table in the supplied booklet. Again in part (a)(ii), answers were usually correct with only a minority of candidates giving $P(R \leq 10)$, $P(R \leq 9)$ or $1 - P(R \leq 9)$ as their answers. Answers to part (a)(iii) were almost always correct with evaluation by formula by far the more common method. As in previous series, part (a)(iv) caused candidates more difficulty, with many uncertain as to how to find $P(5 \leq R \leq 10)$. Whilst most candidates did attempt to subtract two cumulative probabilities, they often selected one, or even two, incorrect values. In part (b)(i), most candidates showed correctly that either $0.85 + (0.15 \times 0.80)$ or $1 - (0.15 \times 0.20)$ resulted in 0.97. Incorrect reasoning that also led to the given answer often assumed that a second shot was always made.

Although many correct answers were seen to part (b)(ii), the logic involved proved too much of a challenge for many candidates. Whilst many identified the correct model of $B(50, 0.97)$, most then attempted $P(S = 48)$ by formula. Most correct answers appeared to be obtained directly from a calculator's inbuilt cumulative binomial function rather than by $P(S' \leq 2)$ using tables for $B(50, 0.03)$. In answering part (b)(iii), most candidates attempted to use np but then failed to identify the correct values of n and p . As a result, $80 \times 0.80 = 64$ and $(80 \times 0.15) \times 0.80 = 9.6$ were common incorrect answers.

Q15.

As expected, it was rare to find an incorrect completion of the table in part (a). Although most candidates identified the correct probabilities of 0.10 and 0.15 in part (b), many then multiplied these two values instead of adding them, or obtained 0.80 from

$0.70 + 0.65 - 0.55$. Showing justifications of the two statements in part (c) proved too challenging to most candidates, with no attempt or general prose being the most common responses, rather than the necessary numerical justifications. In fact, it was clearly evident from the overall responses that most candidates had little knowledge of, or were confused about, the terms 'mutually exclusive' and 'independent'. Whilst some candidates used $P(W \cap J) = 0.55$ to show non-mutually exclusive, very few indeed achieved any marks for justifying non-independence.

Q16.

With the exception of part (a)(iii), candidates' answers scored many, and often all, of the remaining 10 marks. As expected, it was rare indeed to find an incorrect completion of the table in part (a)(i). Although most candidates identified the correct probabilities of 0.10 and 0.15 in part (a)(ii), almost half then multiplied these two values instead of adding them. A minority obtained 0.80 from $0.70 + 0.65 - 0.55$ but an even smaller number obtained the correct answer of 0.25 from $0.70 + 0.65 - 2 \times 0.55$. Showing justifications of the two statements in part (a)(iii) proved too challenging to many candidates, with no attempt or general prose being the most common responses rather than the necessary numerical justifications. In fact, it was clearly evident from the overall responses that most candidates had little knowledge of, or were confused about, the terms 'mutually exclusive', 'exhaustive' and 'independent'. Whilst a minority of candidates used $P(W \cap J) = 0.55$ to show non-mutually exclusive, very few indeed achieved any marks for justifying non-independence. Consequently, part (b) proved a welcome relief to most candidates with 4 or 5 marks being by far the most common marks. It was very rare to see an incorrect answer to part (b)(i), but in part (b)(ii) some candidates multiplied 2, instead of 3, probabilities and/or introduced combination multipliers of either 2 or 3.

Q17.

Answers to this probability question were much improved over comparable questions on the same topic in recent papers. Although incorrect answers to part (a)(i) were not unusual ($0.15 \times 0.4 + 0.15 \times 0.75 + 0.15 \times 1 = 0.3225$), most candidates scored a large majority of the remaining 6 marks in part (a), with only a small minority failing to consider conditional probability in part (a)(iv). Most candidates then made the correct connection between part (a)(iv) and part (b)(i). It was pleasing to see the significant proportion of candidates who then realised the step for part (b)(ii), with the result that the awarding of full marks in part (b) was not uncommon.

Q18.

Better candidates were able to score at least 9 marks on this question but weaker candidates often failed to score more than 2 or 3 marks. Almost all candidates scored the 2 marks available in part (a)(i). Whilst many candidates also scored the 2 marks in part (a)(ii), a minority over-complicated the request by trying to evaluate 4 terms, although some were eventually successful. For a large proportion of candidates, part (a)(iii) resulted in a loss of 3 of the 4 marks available due to often-correct attempts at $P(C = 2)$ instead of $P(C \geq 2)$.

Here again some candidates over-complicated the request by trying to evaluate 6 terms rather than 3 for $P(C = 2)$ or 7 terms rather than 4 for $P(C \geq 2)$. Answers to part (b)(i) were often correct but the simple method for answering part (b)(ii), as $1 - \text{part (b)(i)}$, was sometimes not recognised; the lengthy alternative method often resulted in an incorrect answer.

Q19.

Whilst a not insignificant proportion of candidates scored full or almost full marks, sometimes with very little evidence of working, many candidates found parts of this question difficult. Most candidates answered part (a)(i) correctly but then in parts (a)(ii) and (iii) often multiplied two probabilities.

Similarly, responses in part (b) often involved multiplying together three probabilities rather than two. However, where better progress was made, candidates often scored 1 mark for each of $((a)(ii) \times 0.9)$ and (0.25×0.7) but not for $((a)(iii) \times 0.3)$ since it required $((a)(iii) - 0.25) \times 0.3$. Rather surprisingly, a number of candidates made virtually no headway in part (a) but then scored full marks in part (b).

Q20.

Whilst a not insignificant proportion of candidates scored full or almost full marks, many candidates found this question difficult; this despite its similarity to a question on a recent paper. Most candidates answered part (a)(i) correctly but then in parts (a)(ii) and (iii) often multiplied two probabilities. Similarly, responses in part (b)(i) often involved multiplying together three probabilities rather than two.

However, where better progress was made, candidates often scored 1 mark for each of $(a)(ii) \times 0.9$ and (0.25×0.7) but not for $((a)(iii) \times 0.3)$ since it required $((a)(iii) - 0.25) \times 0.3$. Such candidates often also scored 1 of the 2 marks available in part (b)(ii) for the use of $(1 - (b)(i))^5$.

Q21.

This relatively standard probability question enabled most candidates to score at least 7, sometimes 9 but rarely 11, marks. Full marks were usually scored in parts (a)(i) and (ii). Many candidates also scored full marks in the remainder of part (a) but it was not unusual

to see an answer based upon $\frac{60 + 32}{160}$ for part (a)(ii) and/or answers based upon $\frac{18}{60}, \frac{48}{160}$ or even $\frac{30}{48}$ for part (a)(iv).

Fully correct answers to part (b) were rare. Although many candidates were able to

construct the expression $\frac{24 \times 56 \times 32}{160^3}$ for 1 mark or the expression $\frac{24 \times 56 \times 32}{160 \times 159 \times 158}$ for 2 marks, few realised that permutations had to be considered. Of those who did, about half used 3 rather than $3! = 6$. The very small number of candidates who based their correct

answer on
$$\frac{\binom{24}{1} \binom{56}{1} \binom{32}{1}}{\binom{160}{3}}$$
 are worthy of a special mention!

Q22.

Answers to part (a) were often fully correct with most candidates capable of applying the general multiplication law and the addition law for mutually exclusive events. Answers to part (b) were much less impressive. Whilst most candidates clearly had some notion of

what was needed, their attempts were only partially correct. All too often such answers

were the evaluation of $\binom{4}{3}(0.68)^3(0.32)$ or $(0.68)^3(0.105)$.

Q23.

Answers to part (a) were very disappointing and it appeared from candidates' answers that most had no knowledge of probability beyond the application of multiplication laws. However, many candidates scored well in parts (b), (c) and (d), especially as part (c) did require use of the multiplication law (for dependent events). Most candidates scored the mark for part (a)(i) but even here incorrect answers were not a rare event. Answers to parts (a)(ii) & (iii) invariably involved the multiplication law for independent events:

$$(ii) \quad P(G \text{ but not } S) = P(G) \times P(\text{not } S) = 0.315.$$

$$(iii) \quad P(G \text{ or } S \text{ only}) = (ii) + P(\text{not } S) \times P(G) = 0.48.$$

In fact, as realised by some candidates, who were by no means 'high-flyers', the answers required only simple logic as follows:

Since $P(S \text{ or } G \text{ or both}) = 1$ (Andrew uses no other facilities):

$$(ii) \quad P(G \text{ but not } S) = 1 - P(S) = 1 - 0.55 = 0.45;$$

$$(iii) \quad P(G \text{ or } S \text{ only}) = 1 - P(G \& S) = 1 - 0.25 = 0.75.$$

Most candidates realised that the answer to part (b) required $\{(a)(i)\}^4$ and so scored 1 or 2 marks; some had trouble with the number of leading zeros in 0.3^4 . A small minority of candidates attempted to consider the number of days in a month. As mentioned above, most candidates applied the correct laws in part (c) and many gained all 3 marks. This was often followed by the correct answer to part (d) although some candidates calculated $(1 - 0.35 \times 0.45)$ or $(1 - 0.35) \times (1 - 0.45)$.

Q24.

In part (a), most candidates realised that $B(50, 0.08)$ probabilities were required and so used the booklet provided. However, a significant proportion evaluated $P(X \geq x)$ as $1 - P(X \leq x)$ rather than $1 - P(X \leq x - 1)$ and so obtained incorrect answers. In part (b)(i), many candidates obtained correctly 0.684 from $(0.975)^{15}$ but then, for some unknown reason, some attempted part (b)(ii) as $1 - P(X \leq 1)$ by interpolation from $B(15, 0.02)$ and $B(15, 0.03)$, when all that was needed was $1 - 0.684 = 0.316$.

In part (c), many candidates recognised the need for $\{(a)(ii) \times (b)(i)\}$ but not for $\{(a)(i) \times (b)(ii)\}$. However, it was most pleasing to see that a significant number of candidates were able to score full marks on this question.

Q25.

This was perhaps the best answered question on the paper, with most candidates well prepared for this type of probability question. Almost all candidates scored at least 4 marks, most scored at least 6 marks and many achieved the full 8 marks. Answers to part (a) were almost always correct with simple arithmetic slips the usual cause of a lost mark. Again in part (b), many candidates scored full marks. Those that did not either forgot to

add in their answer to part (a) or made a numerical slip in the two new terms. In part (c), full marks were much less in evidence. In the main this was due to not fully understanding the phrase at “least one” and thereby omitting $P(\text{one})$ or $P(\text{three})$. Those candidates who realised that $P(\text{at least one}) = 1 - P(\text{none})$ invariably scored all 3 marks.

Q26.

This was the best answered question on the paper and centres are to be congratulated on their preparation of candidates for this type of probability question. Almost all candidates scored at least 6 marks, most scored at least 8 marks and many achieved the full 10 marks. Answers to parts (a) and (b) were almost always correct with simple arithmetic slips the usual cause of lost marks. Again in part (c), most candidates scored full marks. Those that did not give an answer of ‘answer (b) $\times$ 3’, had a numerical error in one of their three possibilities or multiplied the latter together, rather than adding them. In part (d), full marks were again the norm. When this was not the case, the usual error was failing to add $P(\text{one})$ (= answer (c)) to a correct value for $P(\text{two})$.

Q27.

This was a very high scoring question with many candidates scoring full marks. Only part (b)(ii) presented any noteworthy difficulty where $0.2 + 0.25 - 0.18$ was the most common incorrect approach.

Q28.

Overall this was the best answered question on the paper with more than 65% of candidates achieving full marks; a credit to centres’ teaching of Bayes’ Theorem. On the rare occasions when full marks were not scored, it was generally in parts (a)(iii) and/or (b) for using other than Bayes’ Theorem.

Q29.

This was probably the best answered question on the paper with many candidates, often through first constructing a tree diagram, scoring full marks. It was thus rare to see candidates not scoring the full 7 marks in part (a). When marks were lost, it tended to be in part (a)(ii). Part (b) proved a good discriminator for the more able candidates. Many candidates recognised the need for $p_1^2 \times p_2$ many fewer included the multiplier of 3 and/or had correct values for both p_1 and p_2 . A small minority of candidates attempted to apply a binomial model.

Q30.

Many candidates scored full or almost full marks on this probability question, sometimes even those who scored very little elsewhere on the paper. Few candidates failed to score the first 2 marks but, in part (a)(ii), it was not uncommon to see 0.2×0.4 . In part (b)(i), most candidates scored all 3 marks. However, in part (b)(ii), only 2 of the correct 3 permutations were often used.

Q31.

In this question part (a) was very well answered, most candidates succeeding in obtaining the printed answer correctly. The conditional probability in part (b) was found correctly by a good number of candidates, but others substituted numbers very inappropriately into a

formula and obtained no credit in this part of the question.

Q32.

Most candidates showed a good grasp of basic probability in part (a) of this question. Some incorrectly used subtraction from 1 in part (a)(ii) and some misunderstood the phrase 'exactly one', confusing it with 'at least one', in part (a)(iii). A small proportion of candidates appeared to be confused by the use of percentages in the question.

A good number of candidates were able to earn at least one mark in part (b) by using their answer to part (a)(iii) as the denominator of a fraction, though the numerator was less often correct.

Q33.

Part (a) of this question offered a good number of marks to most candidates. A direct approach using the numbers in the table was more successful than a slavish dependence on formulae, to some extent in part (a)(iii) but even more so in part (a)(v). In this latter part the formula relating to the union of two sets was often used without excluding the intersection as required.

The same considerations applied in part (b)(i), where the use of a formula for conditional probability often led to incorrect calculations, whereas counting the number of British players and British goalkeepers usually gave rise to the correct answer.

In part (b)(ii) most candidates earned no marks. The comments were often related to general footballing considerations rather than to the 30 players referred to in the question, or candidates confused the word 'independent' with the phrase 'mutually exclusive'. It was hoped that the candidates would see that the original probability of the chosen player being a goalkeeper was altered given the information that the chosen player was British, so that the given events could not be independent.

Q34.

The majority of candidates drew a tree diagram here and were then able to write down their results confidently. After the two multiplications in parts (a)(i) and (a)(ii), it was quite common to see a third multiplication in part (a)(iii) instead of an addition of the first two answers. Conditional probability continues to be a source of difficulty for many candidates, but there were several instances of good work in part (b) from candidates who had trouble with other questions on the paper.

Q35.

The performance of the candidates on this question was, as often with questions on probability, of variable quality. Most candidates seemed to find parts (a)(i) and (a)(ii) straightforward, but the request in part (a)(iii) was not well answered - a common mistake being to consider the probability of 'exactly one' successful outcome instead of 'at least one' successful outcome.

This, in turn, caused some loss of marks in part (b), where most candidates showed no true understanding of conditional probability. Some candidates found themselves trying to divide 0.9 by 0.66. The examiners could see how this had arisen, but no credit could be given for a value greater than 1 for a probability.